Word Problems & Reasoning

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Number and digit reasoning**

A number decreased by 9 equals 34. Find the number.

$x - 9 = 34 \Rightarrow x = 43$.

2 **[Foundational]** • **Target: Number and digit reasoning**

Three times a number, minus 7, equals 32. Find the number.

$3x - 7 = 32 \Rightarrow 3x = 39 \Rightarrow x = 13$.

3 **[Foundational]** • **Target: Number and digit reasoning**

The sum of two numbers is 58. One number is 12 more than the other. Find both numbers.

Let the smaller be x: $x + (x + 12) = 58 \Rightarrow 2x = 46 \Rightarrow x = 23$. The numbers are 23 and 35.

4 **[Foundational]** • **Target: Number and digit reasoning**

A two-digit number has digit sum 11. Reversing its digits gives a number 45 more than the original. Find the original number.

Let the digits be t (tens) and u (units): $t + u = 11$ and $(10u + t) - (10t + u) = 45 \Rightarrow u - t = 5$. Solving together: $u = 8$, $t = 3$. The original number is 38.

5 **[Foundational]** • **Target: Age problems**

A father is 4 times as old as his daughter. In 5 years, he will be 3 times as old as she will be. Find their current ages.

Let the daughter be x, father $4x$: $4x + 5 = 3(x + 5) \Rightarrow 4x + 5 = 3x + 15 \Rightarrow x = 10$. Daughter is 10, father is 40.

6 **[Foundational]** • **Target: Age problems**

Sara is twice as old as her brother. In 6 years, the sum of their ages will be 42. Find their current ages.

Let the brother be x, Sara $2x$: $(x + 6) + (2x + 6) = 42 \Rightarrow 3x = 24 \Rightarrow x = 10$. Brother is 10, Sara is 20.

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7 [Foundational] • Target: Age problems

Ahmed is 5 years older than Salem. Seven years ago, Ahmed was twice as old as Salem. Find their current ages.

Let Salem be x, Ahmed $x+5$: $(x+5-7)=2(x-7) \Rightarrow x-2=2x-14 \Rightarrow x=12$. Salem is 12, Ahmed is 17.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Age problems

The sum of a mother's and daughter's ages is 50. The mother is 20 years older than the daughter. Find both ages.

Let the daughter be d: $d+(d+20)=50 \Rightarrow 2d=30 \Rightarrow d=15$. Daughter is 15, mother is 35.

9 [Intermediate] • Target: Distance, rate and time

A train travels 180 km in 2 hours. At the same speed, how far does it travel in 5 hours?

Speed $=180 \div 2=90$ km/h. Distance $=90 \times 5=450$ km.

10 [Intermediate] • Target: Distance, rate and time

Two cities are 350 km apart. A car leaves one city for the other at 70 km/h. How long does the trip take?

Time $=350 \div 70=5$ hours.

11 [Intermediate] • Target: Distance, rate and time

Two cyclists start at the same point and ride in opposite directions, at 15 km/h and 20 km/h. After how many hours will they be 105 km apart?

Combined speed $=15+20=35$ km/h. Time $=105 \div 35=3$ hours.

12 [Intermediate] • Target: Distance, rate and time

A plane flies against a headwind at 480 km/h and covers 1,440 km. How many hours does the flight take?

Time $=1{,}440 \div 480=3$ hours.

13 [Intermediate] • Target: Work-rate problems

Worker A can paint a room in 6 hours; Worker B can paint the same room in 3 hours. Working together, how long will it take them?

Combined rate $=\frac{1}{6}+\frac{1}{3}=\frac{1}{6}+\frac{2}{6}=\frac{3}{6}=\frac{1}{2}$ room/hour. Time $=2$ hours.

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14 [Intermediate] • Target: Work-rate problems

A pipe fills a tank in 10 hours; a second pipe fills it in 15 hours. Working together, how long to fill the tank?

Combined rate $= \frac{1}{10} + \frac{1}{15} = \frac{3}{30} + \frac{2}{30} = \frac{5}{30} = \frac{1}{6}$. Time $= 6$ hours.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Work-rate problems • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

6 workers can complete a job in 8 days. How many days would 4 workers take, at the same rate?

Total work $= 6 \times 8 = 48$ worker-days. With 4 workers: $48 \div 4 = 12$ days.

16 [Challenging] • Target: Work-rate problems • Cross-Lesson: 3.01 Algebra Essentials

A pipe fills a tank in 12 hours, but a leak can empty a full tank in 24 hours. If both act at once, how long to fill the tank?

Net rate $= \frac{1}{12} - \frac{1}{24} = \frac{2}{24} - \frac{1}{24} = \frac{1}{24}$. Time $= 24$ hours.

17 [Challenging] • Target: Mixture and money problems • Cross-Lesson: 4.01 Geometry & Measurement

A jeweler mixes gold worth 200 riyals/gram with silver worth 50 riyals/gram to make 10 grams of a mixture worth 110 riyals/gram. How many grams of gold are used?

Let g be grams of gold: $200g + 50(10 - g) = 110 \times 10 \Rightarrow 150g = 600 \Rightarrow g = 4$ grams.

18 [Challenging] • Target: Mixture and money problems • Cross-Lesson: 5.01 Quantitative Comparisons

A shopkeeper has 15 riyals in 1-riyal and 2-riyal coins, 10 coins in total. How many of each coin does she have?

Let x be 1-riyal coins, y be 2-riyal coins: $x + y = 10$, $x + 2y = 15$. Subtracting: $y = 5$, $x = 5$.

19 [Challenging] • Target: Mixture and money problems • Cross-Lesson: 6.01 Sequences & Patterns

A solution is 40% acid. How many liters of pure water must be added to 20 liters of this solution to dilute it to 25% acid?

Acid amount stays $0.4 \times 20 = 8$ liters. $\frac{8}{20 + w} = 0.25 \Rightarrow 20 + w = 32 \Rightarrow w = 12$ liters.

20 [Challenging] • Target: Mixture and money problems • Cross-Lesson: 7.01 Data Analysis & Probability

Event tickets cost 30 riyals for adults and 18 riyals for children. 150 tickets were sold for 3,780 riyals in total. How many adult tickets were sold?

Let a be adults, so $150 - a$ children: $30a + 18(150 - a) = 3{,}780 \Rightarrow 12a = 1{,}080 \Rightarrow a = 90$ adult tickets.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Number and digit reasoning

Four times a number, minus 9, equals 47. Find the number.

a. 56

b. 14 ✓

c. 11

d. 9.5

$4x - 9 = 47 \Rightarrow 4x = 56 \Rightarrow x = 14$.

22 [Foundational] • Target: MCQ — Number and digit reasoning

The sum of two numbers is 74. One number is 18 more than the other. Find the larger number.

a. 46 ✓

b. 37

c. 28

d. 56

$x + (x + 18) = 74 \Rightarrow x = 28$; larger $= 46$.

23 [Foundational] • Target: MCQ — Number and digit reasoning

A number is increased by 25% to give 60. Find the original number.

a. 75

b. 40

c. 45

d. 48 ✓

$1.25x = 60 \Rightarrow x = 48$.

24 [Foundational] • Target: MCQ — Number and digit reasoning

A number plus twice itself equals 27. Find the number.

a. 13.5

b. 27

c. 9 ✓

d. 18

$x + 2x = 27 \Rightarrow 3x = 27 \Rightarrow x = 9$.

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25 [Foundational] • Target: MCQ — Age problems

A father is 5 times as old as his son. In 4 years, he will be 3 times as old as his son will be. Find the father's current age.

a. 20 ✓

b. 16

c. 24

d. 4

$5x + 4 = 3(x + 4) \Rightarrow 2x = 8 \Rightarrow x = 4$; father $= 20$.

26 [Foundational] • Target: MCQ — Age problems

Huda is twice as old as her sister. In 8 years, the sum of their ages will be 46. Find Huda's current age.

a. 10

b. 18

c. 14

d. 20 ✓

$(x + 8) + (2x + 8) = 46 \Rightarrow x = 10$; Huda $= 20$.

27 [Foundational] • Target: MCQ — Age problems

Khalid is 6 years older than Faisal. Five years ago, Khalid was twice as old as Faisal. Find their current ages.

a. Faisal $= 17$, Khalid $= 11$

b. Faisal $= 11$, Khalid $= 6$

c. Faisal $= 5$, Khalid $= 11$

d. **Faisal** $= 11$**, Khalid** $= 17$ ✓

$(x + 6 - 5) = 2(x - 5) \Rightarrow x + 1 = 2x - 10 \Rightarrow x = 11$; Khalid $= 17$.

28 [Foundational] • Target: MCQ — Age problems

The sum of a mother's and son's ages is 54. The mother is 24 years older than the son. Find the son's age.

a. 30

b. 15 ✓

c. 27

d. 39

$x + (x + 24) = 54 \Rightarrow 2x = 30 \Rightarrow x = 15$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: MCQ — Age problems

In 8 years, Mona will be twice as old as she was 4 years ago. Find her current age.

a. 8

b. 20

c. 16 ✓

d. 12

$x + 8 = 2(x - 4) \Rightarrow x + 8 = 2x - 8 \Rightarrow x = 16$.

30 [Intermediate] • Target: MCQ — Age problems

Tariq is currently twice as old as his son. If Tariq is 40, find his son's age.

a. 18

b. 20 ✓

c. 22

d. 80

$40 = 2s \Rightarrow s = 20$.

31 [Intermediate] • Target: MCQ — Distance, rate and time

Two cyclists start at the same point and ride in opposite directions, at 12 km/h and 18 km/h. After 3 hours, how far apart are they?

a. 54

b. 90 ✓

c. 30

d. 36

Combined speed $= 30$ km/h; $30 \times 3 = 90$ km.

32 [Intermediate] • Target: MCQ — Distance, rate and time

A car travels at 90 km/h for 2.5 hours. Find the distance covered.

a. 225 ✓

b. 36

c. 92.5

d. 180

$90 \times 2.5 = 225$ km.

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33 [Intermediate] • Target: MCQ — Work-rate problems

Worker A can paint a room in 8 hours; Worker B can paint the same room in 4 hours. Working together, how long will it take?

a. 4 hours

b. 12 hours

c. 6 hours

d. $\frac{8}{3}$ **hours ✓**

$\frac{1}{8}+\frac{1}{4}=\frac{3}{8}$ per hour; time $=\frac{8}{3}$ hours.

34 [Intermediate] • Target: MCQ — Work-rate problems

Worker C can complete a task in 9 hours. After working alone for 3 hours, what fraction of the task remains?

a. $\frac{2}{3}$ ✓

b. $\frac{1}{3}$

c. $\frac{1}{6}$

d. $\frac{1}{9}$

In 3 hours, $\frac{3}{9}=\frac{1}{3}$ is done; remaining $=\frac{2}{3}$.

35 [Intermediate] • Target: MCQ — Mixture and money problems

A shopkeeper has 21 riyals in 1-riyal and 2-riyal coins, 15 coins in total. How many 2-riyal coins does he have?

a. 9

b. 15

c. 6 ✓

d. 3

$x+y=15, x+2y=21 \Rightarrow y=6$.

36 [Intermediate] • Target: MCQ — Mixture and money problems

A solution is 30% acid. How many liters of pure water must be added to 15 liters of this solution to make it 20% acid?

a. 4.5

b. 7.5 ✓

c. 5

d. 10

$\frac{4.5}{15+w}=0.2 \Rightarrow 15+w=22.5 \Rightarrow w=7.5$.

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37 [Intermediate] • Target: MCQ — Mixture and money problems

Event tickets cost 25 riyals for adults and 15 riyals for children. 200 tickets were sold for 4,300 riyals. How many adult tickets were sold?

a. 150

b. 130 ✓

c. 100

d. 70

$25a + 15(200 - a) = 4{,}300 \Rightarrow 10a = 1{,}300 \Rightarrow a = 130.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Mixture and money problems • Cross-Lesson: 5.01 Quantitative Comparisons

A syrup is 25% sugar. How many kilograms of pure sugar must be added to 8 kg of this syrup to make it 40% sugar?

a. 3.2

b. 4

c. 1

d. 2 ✓

$\frac{2+x}{8+x} = 0.4 \Rightarrow 2 + x = 3.2 + 0.4x \Rightarrow x = 2.$

39 [Challenging] • Target: MCQ — Logical deduction • Cross-Lesson: 6.01 Sequences & Patterns

Omar, Bandar and Nasser have 44 books together. Omar has twice as many as Bandar, and Nasser has 8 more than Bandar. Find how many books Bandar has.

a. 22

b. 17

c. 9 ✓

d. 18

$b + 2b + (b + 8) = 44 \Rightarrow 4b = 36 \Rightarrow b = 9.$

40 [Challenging] • Target: MCQ — Logical deduction • Cross-Lesson: 7.01 Data Analysis & Probability

In a class, every student plays chess or checkers (or both). 20 play chess, 15 play checkers, and 8 play both. How many students are in the class?

a. 27 ✓

b. 35

c. 12

d. 43

$20 + 15 - 8 = 27.$

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41 [Challenging] • Target: MCQ — Logical deduction • Cross-Lesson: 1.01 Arithmetic & Number Sense

Four consecutive odd integers sum to 96. Find the largest one.

a. 30

b. 27 ✓

c. 24

d. 21

$4n + 12 = 96 \Rightarrow n = 21$; largest $= n + 6 = 27$.

42 [Challenging] • Target: MCQ — Logical deduction • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Three consecutive integers sum to 72. Find the smallest one.

a. 25

b. 24

c. 23 ✓

d. 22

$3n + 3 = 72 \Rightarrow n = 23$.

43 [Challenging] • Target: MCQ — Working backward • Cross-Lesson: 3.01 Algebra Essentials

After spending half his money on a book, then 15 riyals on lunch, Faisal has 25 riyals left. How much did he start with?

a. 100

b. 65

c. 80 ✓

d. 40

$\frac{x}{2} - 15 = 25 \Rightarrow \frac{x}{2} = 40 \Rightarrow x = 80$.

44 [Challenging] • Target: MCQ — Working backward • Cross-Lesson: 4.01 Geometry & Measurement

A number is doubled, then 6 is subtracted, giving 20. Find the original number.

a. 16

b. 7

c. 26

d. 13 ✓

$2x - 6 = 20 \Rightarrow x = 13$.

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45 [Challenging] • Target: MCQ — Working backward • Cross-Lesson: 5.01 Quantitative Comparisons

A number is halved, then 10 is added, giving 34. Find the original number.

a. 48 ✓

b. 44

c. 88

d. 24

$\frac{x}{2} + 10 = 34 \Rightarrow \frac{x}{2} = 24 \Rightarrow x = 48.$

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